

```
In [ ]: import re
import random
import string
import textwrap

import numpy as np
import pandas as pd

import warnings
warnings.filterwarnings('ignore')
```

1. Өгөгдсөн хүснэгтээс нийт ялгаатай утгатай элементүүдийн тоог ол.

```
In [ ]: def generate_random_string(length):
    return ''.join(random.choices(string.ascii_letters + string.digits, k = length))

def generate_random_dataframe(n):
    df = pd.DataFrame({'Col1': [generate_random_string(10) for _ in range(n)],
                        'Col2': [generate_random_string(10) for _ in range(n)]})

    return df
```

```
In [ ]: df = generate_random_dataframe(n = 30)

print("First col:")
print(df['Col1'].value_counts().sample(10))

print("\nSecond col:")
print(df['Col2'].value_counts().sample(10))
```

```
First col:
vUpG1P7YX0    1
5v2XSDhiLH    1
gJwjTTK4xq    1
nGqm86mSb2    1
uMXYdKkmd7    1
ItHJRJ2Gq0    1
oNeT6iIdYB    1
PfNRQ6A6YT    1
0205Bh5lxu    1
sFJbArsWCq    1
Name: Col1, dtype: int64
```

```
Second col:
iTo1l3VbWe    1
BLESSzQcsZ    1
3ZaqKAVCox    1
CnYrvhJsiD    1
DJwDDVrt7X    1
EVgy3h3u9g    1
jpRbf6d5cu    1
D02VLuylSa    1
ZzGz4XfBB4    1
BKA8NNZh1l    1
Name: Col2, dtype: int64
```

2. Өгөгдсөн өгүүлбэрийн үг бүр нь хэдэн удаа давтагдсанг ол.

```
In [ ]: lorem_ipsum = """Lorem Ipsum is simply dummy text of the printing and
typesetting industry Lorem Ipsum has been the industry's standard dummy
text ever since the 1500s, when an unknown printer took a galley of type and
scrambled it to make a type specimen book It has survived not only five centuries,
but also the leap into electronic typesetting, remaining essentially unchanged
It was popularised in the 1960s with the release of Letraset sheets containing
Lorem Ipsum passages, and more recently with desktop publishing software
like Aldus PageMaker including versions of Lorem Ipsum."""
```

```
In [ ]: # 1. No puncts
text_no_punct = re.sub(r'[^w\s\']', '', lorem_ipsum)
print(textwrap.fill(text_no_punct))
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry Lorem Ipsum has been the industry's standard dummy text ever since the 1500s when an unknown printer took a galley of type and scrambled it to make a type specimen book It has survived not only five centuries but also the leap into electronic typesetting remaining essentially unchanged It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

```
In [ ]: # 2. 's and s after numbers
text_no_suffix = re.sub(r'(?<=\d)s\b', '', text_no_punct)
print(textwrap.fill(text_no_suffix))
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry Lorem Ipsum has been the industry's standard dummy text ever since the 1500 when an unknown printer took a galley of type and scrambled it to make a type specimen book It has survived not only five centuries but also the leap into electronic typesetting remaining essentially unchanged It was popularised in the 1960 with the release of Letraset sheets containing Lorem Ipsum passages and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

```
In [ ]: # 3. Remove ' letters
text_no_apos = re.sub(r'\b[w+]\w+\b', '', text_no_suffix)
print(textwrap.fill(text_no_apos))
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry Lorem Ipsum has been the standard dummy text ever since the 1500 when an unknown printer took a galley of type and scrambled it to make a type specimen book It has survived not only five centuries but also the leap into electronic typesetting remaining essentially unchanged It was popularised in the 1960 with the release of Letraset sheets containing Lorem Ipsum passages and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

```
In [ ]: # 4. contractions like e.g 're, 've
text_no_contractions = re.sub(r'\b\w+\'(?:re|ve|s|m|d|ll)\b', '', text_no_apos)
print(textwrap.fill(text_no_contractions))
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry Lorem Ipsum has been the standard dummy text ever since the 1500 when an unknown printer took a galley of type and scrambled it to make a type specimen book It has survived not only five centuries but also the leap into electronic typesetting remaining essentially unchanged It was popularised in the 1960 with the release of Letraset sheets containing Lorem Ipsum passages and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

```
In [ ]: # 5. No extra space
clean_text = re.sub(r'\s+', ' ', text_no_contractions).strip()
print(textwrap.fill(clean_text))
```

Lorem Ipsum is simply dummy text of the printing and typesetting industry Lorem Ipsum has been the standard dummy text ever since the 1500 when an unknown printer took a galley of type and scrambled it to make a type specimen book It has survived not only five centuries but also the leap into electronic typesetting remaining essentially unchanged It was popularised in the 1960 with the release of Letraset sheets containing Lorem Ipsum passages and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum

```
In [ ]: word_cnt = pd.DataFrame({"words": [word.strip() for word in clean_text.split()]})

print("Each word cnt in sentence:")
print(word_cnt['words'].value_counts())
```

Each word cnt in sentence:

the	6
Lorem	4
of	4
Ipsum	4
and	3
..	
book	1
not	1
only	1
five	1
versions	1

Name: words, Length: 66, dtype: int64

3. Өгөгдсөн хүснэгтээс 'хүз' тэмдэгт мөрийг агуулсан тэмдэгт мөрүүдийг хэвлэ.

```
In [ ]: def generate_random_string(length):
    prefix_length = random.randint(0, length - 3)
    suffix_length = length - prefix_length - 3

    xyz_pattern = 'xyz'
    prefix = ''.join(random.choices(string.ascii_letters + string.digits, k = prefix_length))
    suffix = ''.join(random.choices(string.ascii_letters + string.digits, k = suffix_length))

    random_number = np.random.randint(6)
    if random_number % 3 == 0:
        return prefix + xyz_pattern + suffix
    return prefix + suffix
```

```
In [ ]: xyz_objects = pd.DataFrame({'Col': [generate_random_string(15) for _ in range(100)]})
xyz_objects.sample(15)
```

Out[]:

	Col
20	B6mxyz9ssKnYhRj
24	jGkG6Mm6VPwl
8	fGpbSWcKaW6a
55	X8jMYxOIH9ZG
53	0zmLjReehsCc
63	7qnY6Xp7Smv9
52	lCFj9PxyzVC2g4e
17	xmHN1YPXq52L
6	lsfbVkWgAh63
81	Ypl0TuyRBxCR
64	x1Ki8biqRhl5
92	TlkDPLcd1xyzJrJ
13	06sHxdnXcxSg
32	2WznOZ86qRvO
1	M8ugxyz1j3e58He

```
In [ ]: print("Column values containing xyz:")
print(xyz_objects[xyz_objects['Col'].str.contains('xyz')].reset_index(drop = True))
```

Column values containing xyz:

	Col
0	M8ugxyz1j3e58He
1	axyzULXn4tBb3hb
2	y9zkmxyzL3IG7Do
3	5MUqok3xyzXBju0
4	Jizxyz9xq2dWaQA
5	yxyzyvhWC0zAR9g
6	94WnNt0rQ1xyzbx
7	B6mxyz9ssKnYhRj
8	kX9i160mPq4yxyz
9	7jbxyz9xIaywEVk
10	2bX6FCDXuxyziDJ
11	YjnZhc7d0xyzN7o
12	LxyzQkdcIc1BitM
13	nxyzAwPSNi59yvh
14	qMIBxyz1yDna3fA
15	9oukzV7wuxyzamI
16	hZaIxyzlhnc0BQM
17	lCFj9PxyzVC2g4e
18	92s1lw15zv9xyzB
19	NRxyz8n6ogcQwZ
20	piaUqiq1xyzsKyM
21	zafadg6JxyzYfCJ
22	TxyzBSFjhB2d5RE
23	hjQxyzJXNAJQjee
24	i5U5xRbl0Vaexyz
25	2tWp6dCxyzh2io1
26	0HqGuxyzkDx730B
27	1w4Mpo9xyzijkBZ
28	22W8xyzo94p03eb
29	1Tlo0rpjxyzkpww
30	TlkDPLcd1xyzJrJ
31	PCENVAW0soYAxyz
32	gcft5Hzx1f2xyzG

**4. Өгөгдсөн хүснэгтээс 'хуз' тэмдэгт мөрөөр эхэлсэн эсвэл
төгссөн тэмдэгт мөрүүдийг хэвлэ.**

```
In [ ]: xyz_objects = pd.DataFrame({'Col': [generate_random_string(15) for _ in range(1000)]})
xyz_objects.sample(15)
```

```
Out[ ]:
```

	Col
576	KVegJvxhoLgv
623	eNVhgOGwB4xyz5l
264	duHLj8gKFGE1
945	zXOPS4OvS9AD
323	vhmKmlxyz15PSRS
654	lFbWd0VSWZjz
190	ZAbis1KW98Q8
238	bAEdZ3K8Dokj
666	3uKXZ9AxyzzSN8A
946	a7VR5Uha9xxt
482	fSmPUbPAUrsa
829	xyzlPN2sgrZZikx
693	QUkvvebxyzsk1le
356	1u0J2xyz70VoT0r
746	PHDjFN8u6CB2

```
In [ ]: print("Column values startswith or endswith:")
print(xyz_objects[(xyz_objects['Col'].str.endswith('xyz')) |
                  (xyz_objects['Col'].str.startswith('xyz'))].reset_index(drop = True))
```

Column values startswith or endswith:

	Col
0	xyzWYur14kaeAAb
1	oFqHj0C5L2ldxyz
2	BGq9NQbWtxGvxyz
3	dQ6T97grwitZxyz
4	xyze37H1yKWCgX4
5	xyzMTuCB4WHbIop
6	PlzWseMnE4klxyz
7	wyQSESwSbD5gxyz
8	xyzDtVQEIDaeMCQ
9	xyzputdmecrpmYJ
10	xyzn1iR02rFNQ8S
11	1ckCLtjbtdVQxyz
12	bI48HQLSZUjAxyz
13	LHMo0aoMg1fCxyz
14	xyzYZgEB2K2dEF4
15	J3VEXqsy4FZ7xyz
16	03HY4IWofPPGxyz
17	xyzzEmQJePyNusj
18	xyzuD70kRkTmz8V
19	Ra3tp5EFxscAxyz
20	sFhyH3mk2oJRxyz
21	rz9GH1na0baVxyz
22	xyztX2k7AKoikAP
23	E3SiWyxlreuexyz
24	xyzyP907vDqbklC
25	xyzjXRS3udY3MX4
26	lpvPXS0wEwHaxyz
27	WtoNBcx3cFLYxyz
28	xyz9Du043n0Tdpx
29	gNrf0Glnvll0xyz
30	2HSELRMGr1c2xyz
31	9nr9u1UpKeYdxyz
32	h0bq1BbmZYh9xyz
33	xyz488SvuDLgMYJ
34	xyzMabTuBd9Lnqy
35	xyzKW1D1IQ6PvLf
36	xyzIKToHiijQM8s
37	xyzGv005zHPqt23
38	l0h0ILfwoTnfxyz

```
39 2NWBt4k44mMXxyz
40 tLVr0mMZc00hxyz
41 zrQa8m3Lmypkxyz
42 xyzMPvBPYU94tMo
43 xyzlPN2sgrZZikx
44 xyzSqjm1AN8k8NE
45 xyzzp0bzUy7M5hF
46 xyzsGmxV7nStRFn
47 yXkF4wpB6ZgYxyz
48 id7c6lfi1crGxyz
49 xyzSJRZbM6R6arK
```

5. Хамгийн эхний анхны 50 тоог олох функц бич.

```
In [ ]: def sieve(n):
        primes = []
        is_prime = [True] * (n * 10)

        for num in range(2, int(n * (n**0.5)) + 1):
            if is_prime[num]:
                primes.append(num)
                for multiple in range(num * num, n * 10, num):
                    is_prime[multiple] = False

        for num in range(max(2, int(n * (n**0.5)) + 1), n * 10):
            if is_prime[num]:
                primes.append(num)

        return primes[:n]
```

```
In [ ]: print("First 50 primes:")
        print(textwrap.fill(', '.join(str(num) for num in sieve(50))))
```

```
First 50 primes:
2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61,
67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137,
139, 149, 151, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211,
223, 227, 229
```

